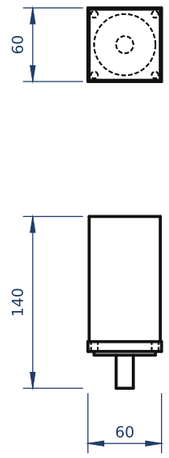
ENVELOPE (as the model draws it) 1:5



specification	value
motor	Nanotec / SMD APBA60M048030-E: 200 W, 48 V, 3000 rpm, 0.64 N·m rated, 1.92 N·m peak, 6 A, IP65, 2500 CPR encoder, 60 mm flange (NEMA 24), body 110 mm, about 1 kg (smd.ee; nanotec.com)
duty	0.94 N·m at the belt, 61 W at its shaft (38 % of rated at 2400 rpm), 76 W electrical; run-up 32 s at its rated torque (§4)
driver	a 48 V BLDC driver with a speed loop, ramps and a current limit at 1.5 × rated (the design's torque fuse, §4.3): Nanotec's matching driver (to choose)
supply	48 V DC, 300 W
alternates	Oriental Motor BLV series 200 W (48 V DC driver); Nanotec DB80S048030 (283 W); a 0.25 kW 4-pole induction motor on a VFD, 24T / 56T (the economy line, §6)
envelope	the flange's PCD Ø70, the spigot Ø50 and the shaft Ø14 × 30 keyed are the 60 mm frame's class [IR]: confirm on the maker's drawing

NOTES

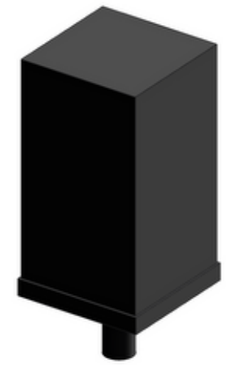
1 The motor's flange stands on the motor plate (DRV-121), its shaft down into the 20T pulley.

2 Datasheets: smd.ee/products/dc-brushless-motors/apba60m048030-e; nanotec.com 2881-apba60m048030-e.

REVISIONS

Rev	Date	Description
A	2026-10-11	First issue, draft for quotation

status	BOUGHT · for the drive's first cut (PROPOSED)
qty per machine	1
used in	DRV-120
source	docs/drive-gear-belt.md §6



3-D view (axonometric, not to scale)

DCCREG turbine · bought parts

Motor, BLDC 200 W 48 V (APBA60M048030-E), with driver

Drawing no. **DCCREG-P-015**

Material: brushless DC motor, 60 mm flange, encoder

Mass: 1 kg	Qty per machine: 1	Rev A Sheet 99 of 103 A3
Scale: 1:5	Tolerances: ISO 2768-mK	Date: 2026-10-11
Drawn: generated by docs/make_drawing_set.py	Checked: — Approved: —	**BOUGHT · SOURCE CONTROL**